

Towards Neural Cellular Automata for Spectral Data Simulation of Symptomatic Crop Leaves

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Abstract. Crop pathogens pose considerable risks to global food production, resulting in substantial yield losses, yet current plant health monitoring still largely relies on manual assessments. Automated disease detection using artificial intelligence (AI) integrated into modern multispectral sensor systems offers spatio-temporally highly-resolved information beyond the visible spectrum, enabling earlier pathogen detection and enhanced decision support in farming practices. However, robust AI-based recognition requires extensive and diverse training datasets covering various disease stages, which are costly and labour-intensive to acquire because they entail crop cultivation, pathogen inoculation, and multispectral image acquisition. This paper proposes Neural Cellular Automata (NCA) variants as a parameter-efficient approach for multispectral crop leaf image simulation. As a proof-of-principle, the NCA is trained to reconstruct fixed target leaf sections of 50×50 pixels across 5, 10, 15, and 20 spectral bands. In a second in silico experiment, the general capability of NCAs to simulate the transition from healthy to pathogen-induced symptomatic tissue across distinct disease stages is explored. Quantitative evaluation demonstrates structural fidelity and spectral fidelity using RMSE, SSIM, and SAM metrics, respectively. Moreover, parameter counts of no more than 18K confirm the computational efficiency of our approach. Further qualitative assessment corroborates the hypothesized potential of NCAs for simulating spectral images of diseased crops and in turn for contributing the mitigation of data scarcity in plant disease detection.

Keywords: Digital Agricultural Technology · Data Augmentation · Spectral Imaging · Self-Organizing Systems · Pathogen-Monitoring

1 Introduction

Current agricultural systems face considerable challenges in meeting the food security and biodiversity [2] requirements outlined by the UN’s sustainability goals. Contemporary digital technologies, including *artificial intelligence* (AI), improve said agricultural innovations [6]. Pests and pathogens are estimated to be responsible for considerable yield losses of up to 40% in major crops [25, 31]. Early and reliable detection of crop diseases is critical to allow for appropriately timed agricultural measures and prevent significant yield loss and quality reduction. Such evaluations of crop diseases are primarily monitored through manual,

sometimes destructive, random assessments of the crop stand carried out by experts. Automated pathogen detection using *artificial intelligence* (AI) embedded in state-of-the-art sensor-actuator systems has the potential to overcome the limitations of manual disease management procedures. In this regard, contemporary hyperspectral imaging technologies provide information beyond the visible light spectrum, enabling more reliable and non-destructive early-stage recognition of pathogen-induced crop diseases [13]. However, implementing a robust disease detection using AI-based methods requires extensive, systematically prepared training data sets. To this end, cultivated crops must be manually inoculated with relevant pathogens in extensive laboratory and greenhouse experiments, which however do not reflect in-situ field conditions. Subsequent images must be acquired using spectral imaging techniques. Systematically evaluating and annotating such data is complex due to the nonlinear progression of the disease [5].

Generative AI models can synthesize realistic RGB images and are being used to augment training data in agricultural domains [23, 24]. These models approximate the statistical properties of real data and can synthesize new images that corresponds to the original distribution. Nonetheless, state-of-the-art models are reliant upon a substantial quantity of training data, especially in highly specialized domains. Given the computational and resource-intensive nature of these models, even when constrained to the synthesize of RGB images, this underscores the need for approaches simulating spectral data in a computationally and data-efficient manner.

In this context, *neural cellular automata* (NCA) [16], offer promising potential for simulating higher-dimensional image data beyond RGB images. NCAs as self-organizing systems, rely on a single-cell model architecture, inspired by the cell communication in real organisms. The interaction between those cells is limited to only their immediate neighbors, resulting parameter-light model sizes, while efficiently encoding information [9].

It is hypothesized that employing self-organizing systems, such as NCA-based models, can enable simulating spectral data in highly specialized domains, such as for augmenting plant pathogen detection.

Contribution This paper for the first time explores NCAs as parameter-efficient self-organizing models for reconstructing spectral images of fixed target sections of both healthy and of pathogen induced crop leaves. We aim at providing a first proof-of-principle of the capacity of NCA variants for parameter-light simulation of real-world spectral image data.

The paper structure is as follows: After reviewing related work (cf. Sect. 2), the proposed approach of using NCA variants is described in Section 3. The presented approach is separated in two distinct scenarios: 1.) First up-scaling the spectral resolution from RGB to multispectral images. 2.) Implementing a *signal-responding* NCA variant [27] to simulate the transition from healthy to pathogen-induced symptomatic leaf tissue across distinct disease stages. The results of quantitatively assessing the structural and spectral fidelity, as well as the model parameter counts, of the conducted experiments are presented in are

presented in Section 4, along with exemplary pseudo-RGB and grayscale images of the simulated leaf sections compared to the target leaf sections.

2 Related Work

Agricultural applications of generative models Multiple generative AI models have been employed to synthesize RGB image data [3, 5, 11] and multispectral image data of pathogen-induced crop leaves with five distinct spectral bands [12]. However, current state-of-the-art generative models, such as *denoising diffusion probabilistic models* (DDPMs) [8] face inherent trade-offs regarding their computational efficiency, due to their model size of up to hundreds of millions of trainable parameters [9]. This restricts their use in environments where powerful hardware is limited, such as in the agricultural context. This emphasizes the need for considering computationally efficient models, such as NCAs, as explored in this paper, relying on a decentralized, parameter-light model architectures.

Neural Cellular Automata Within the domain of computational models, *self-organizing systems* consist of simple components, such as cells or agents, that interact through local rules within intermediate neighborhoods, resulting in complex *emergent* and *decentralized* behavior - mimicking phenomena found in nature. *Cellular automata* (CA) [17, 34] are dynamic systems consisting of a set of cells arranged on a predefined grid, known as *world state*. Within the world state, each cell is governed by three main components: 1.) *cell state* - the finite states a cell can take on, 2.) *neighborhood* - the neighboring cells affecting its state, 3.) *update rule* - a locally defined transition from one state to another. Taking into account all three components, at each discrete time step t , every cell x updates its state $s_t(x) \in S$ by applying the update rule F to its own state and those of its neighbors $\{x_0, x_1, \dots, x_{n-1}\}$, yielding the next state $s_{t+1}(x)$. This parallel process [26], identical for all cells, is defined as $s_{t+1}(x) = F(s_t(x), s_t(x_0), s_t(x_1), \dots, s_t(x_{n-1}))$.

Building on these discrete foundations, NCAs integrate a shallow neural network (NN) enabling continuous state-transitions, and resulting in the data-driven discovery of differentiable update rules. Depending on their training strategy, the models have shown to display different behaviors and capabilities: *Growing* a predefined shape, *persisting* a shape as stable organism over time, and *regenerating* after inflicted external damage [4, 16]. NCAs have proven themselves as compact, self-organizing image simulators [16, 28] and have been extended to the application of texture and pattern creation [18, 20–22]. Additional extensions show that, NCAs can be trained to respond to both internal (*genome-coded*) and internal (*environmental*) signals, by implementing the signal to a single cell within the world state for a discrete timestep. This *signal-conditioned* variant is able to respond, and cause one organism to grow into multiple distinct shapes (internal signals), and change in color of grown organisms (external signals) [27]. In addition, overcoming the limitation to low-resolution grid sizes, recent work combines the general NCA variant with a Local Pattern Producing Networks

(LPPN) enabling the simulation of high-resolution images [19]. The general NCA variants discussed above aim to grow predefined RGB images (emojis), or textures, while maintaining persistence and regenerative capabilities. Hybrid-architectures combine decentralized NCA principles with generative modeling, such as diffusion models, synthesizing images of size 512×512 with just 336 K learning parameter counts [10]. Beyond theoretical applications, real-world implementations include high-resolution medical image generation [7]. Collectively, these recent developments underscore growing interest in NCAs and related hybrid approaches for generative modeling, enabling novel applications, ranging from theoretical studies to more practical use cases, such as data augmentation.

This work extends the use of general NCAs variants [16, 27] from theoretical to real-world applications by extending the reconstruction from RGB images to spectral data and simulating pathogen-induced disease progression on crop leaves.

3 Approach

The proposed method focuses on the overarching idea of up-scaling NCA variants initially investigated on toy-like problems [16, 27] for simulating multispectral data of crop leaves. The objective is to investigate their general viability, extending from the visible range (RGB images) to invisible spectral ranges, and thus to multispectral target images. The conducted experiments (cf. Sect. 4) explore *parameter-efficient* NCA architectures by systematic extension from RGB to multiple higher spectral resolutions. The general viability of NCAs is investigated in two distinct scenarios:

1) *Morphogenesis*: A *general* NCA is trained on two given target leaf sections (healthy and pathogen-induced) and expected to grow this fixed target from a single cell. Considering the architecture presented in [16], the aim is to evaluate the models capacity for growing *persistent* targets, in this first experiment, we systematically extended the spectral resolution to multispectral data; this includes 5 and 10 spectral bands to reflect the configurations of state-of-the-art multispectral cameras, as well as an expansion to 15 and 20 spectral bands to evaluate further scalability towards hyperspectral sensors.

2) *Signal-responding virtual inoculation*: The NCA is extended by an adaptation of a *signal-responding* variant [27]. The objective is to enable a transition from a healthy leaf to two distinct pathogen-induced stages, as a first step to a virtual inoculation of digital representations of healthy plant leaves. While relying on fixed regularity captured laboratory image data acquired from the same leaf at two distinct days after inoculation (DAI), the approach is evaluated by conducting the experiment at two different stages of disease severity at a spectral resolution of 10 bands

In the following sections the specific configurations of both the *general* NCA and *signal-responding* NCA will be described more formally:

General In the first scenario, the *general* NCA is adapted to simulate a target leaf section with height H and width W , extending the *world state* to $\mathbb{R}^{H \times W \times C}$. The state of a cell at time step t is defined as a vector of C scalars $\vec{x}_t \in \mathbb{R}^C$, extending the cell grid to a 3D array of shape $\mathbb{R}^{H \times W \times C}$. In detail, $\vec{x}_t = (C_i, C_\alpha, C_h)$, where n *input channels* $C_i := (C_1, \dots, C_n)$ represent the target (e.g., RGB or spectral bands) as continuous values $C_i \in [0, 1]$. The *alpha channel* $C_\alpha \in \mathbb{R}$ differentiates between cell states [16] based on a threshold $l_s = 0.1$: Cells are considered *living* or *mature* if $C_\alpha > l_s$, whereas cells with $C_\alpha \leq l_s$ are considered *growing*. *Hidden channels* $C_h \in \mathbb{R}^h$ do not serve a predefined meaning and can vary in their number h . To evolve a seed into the target, the *update rule* F computes a residual transition added to the current cell state $\vec{x}_{t+1} = \vec{x}_t + \Delta \vec{x}_t$. The intermediate Moore neighborhood $\vec{x}_t(\mathcal{M})$ is convolved with fixed 3×3 filters (Identity, Sobel_x, Sobel_y) to form a perception vector of size $3C$. This vector is processed by a neural network consisting of two 1×1 convolutional layers [16]. The update is applied stochastically, where each cell x independently draws a random value $r_x \sim \mathcal{U}$ at time step t and is updated if $r_x \leq p_{fire}$, with the defined hyperparameter *cell fire rate* $p_{fire} \in [0.1, 0.9]$. Implementing the *persistence* training strategy [16] (i.e., stable organism over time). Batches are sampled from a pool $\mathcal{P}$ of NCA states and evolved for a random number of steps $\sim U\{64, 96\}$. The NN weights are updated during *backpropagation*, applying a cell-wise Mean Squared Error (MSE) loss between the evolved cell grid and the ground truth (GT) target on channels $C_{loss} := C_i, C_\alpha$. To ensure persistence behavior, the highest-loss states in $\mathcal{P}$ are replaced with new cells. (cf. Suppl. Mat. 1.1, 1.2, 1.3 for additional details).

Signal-responding To enable conditional morphogenesis from a healthy leaf into specific pathogen infection stages, the cell state is extended to $\vec{x}_t = (C_i, C_\alpha, C_h, C_g, C_E)$. Internal *genome channels* $C_g \in \{0, 1\}^2$ encode the target (e.g., early or late infection), while a read-only *environment channel* $C_E \in \{0, 1\}$ functions as an external trigger. When $C_E = 0$, The update rule F is trained to remain silent (i.e., suppress updates to C_i) as long as the environmental stimulus remains inactive. This ensures that the healthy leaf state persists. Activating $C_E = 1.0$ at specific coordinates forces the morphological transition dictated by C_g . During training, batches sampled from a pool $\mathcal{P}$ of healthy states are divided equally into a *persistence group* ($C_E = 0$) and a *signaling group* (where $C_E = 1.0$ is applied with a 0.15 probability). After evolution, the loss is computed between evolved individuals and their assigned target, forcing the model to execute morphological changes exclusively when $C_E = 1.0$ holds. To maintain persistent, the highest-loss states in $\mathcal{P}$ are continually replaced with initialized healthy states ($C_E = 0$). (cf. Suppl. Mat. 1.4, 1.5 for additional details).

4 Experiments

Data acquisition The crop data consist of hyperspectral images of soybean (*Glycine max*) and winter wheat (*Triticum aestivum*) leaves collected under

laboratory conditions using the MC023MG-SY-FL 1D VIS hyperspectral line scanner. The pixel values represent the measured absolute *reflectance* and plotted along the range to provide a spectral signature. HSI data was collected in a 24-hour interval time series ranging from one until ten DAI for Asian soybean rust (ASR) (*Phakopsora pachyrhizi*) and from one to 15 DAI for wheat black rust (WBR) (*Puccinia graminis f. sp. tritici*) (cf. Suppl. Mat. 1.6 for additional details regarding the data collection). During preprocessing the plant leaves were manually segmented using GIMP v2.10.38 [29], excluding the image background. The target size for the NCA to simulate the plant tissue was chosen to be 50×50 cells, in both conducted experiments, containing the corresponding target leaf sections, as displayed in Figure 1. In the *first experiment*, both target sections

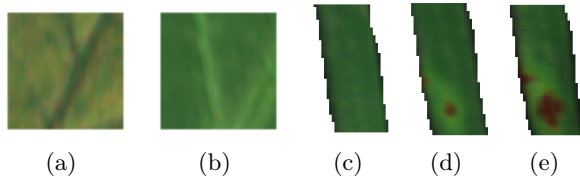


Fig. 1: Target leaf sections as pseudo-RGB for experiment 1: ASR at 8 DAI (a) and healthy soybean (b); and experiment 2: winter wheat with WBR at 1 DAI (c), 6 DAI (d), and 8 DAI (e).

were chosen from different leaves; one from an ASR-infected leaf taken at 8 DAI (Figure 1 a) and one from a healthy soybean leaf (Figure 1b). For the *second experiment*, target sections were acquired from the same winter wheat leaf at consecutive stages. Taking initially healthy state at 1 DAI, and subsequent early symptoms at 6 DAI, and progressively distinct symptoms at 8 DAI (i.e., Figures 1d-e). In the conducted experiments, the HSI data was resampled. In the *first experiment*, the hyperspectral data was resampled to be visualized as *baseline* pseudo-RGB images and to correspond to spectral bands in state-of-the-art multispectral cameras containing either 5 (MicaSense Altum-PT bands) [15] or 10 (MicaSense Dual Camera System) [14] bands, both utilized in agricultural domains. Additionally, the hyperspectral data was resampled to 15 and 20 bands to equally represent spectral information over the full range captured by the employed sensor. In the *second experiment*, the data was resampled specifically to ten multispectral bands, as explained above. For details regarding data preprocessing and exact central wavelengths and bandwidths please cf. Suppl. Mat. 1.7, 1.8. The resampled images were instance-normalized to $[0, 1]$, mapping the lowest and highest values to 0 and 1 respectively, across the reflectance values of either the RGB or multispectral spectral bands, with the reflectance of each band stored as an input channel C_i of the cell state.

Experimental setups The first experiment (*spectral resolution scaling*) evaluates the general NCA variant [16] for generating persistent sections of healthy and

pathogen-induced plant leaves. Starting from RGB as a proof-of-principle, spectral resolution is scaled to 5 and 10 spectral bands, representing state-of-the-art multispectral sensors, and further to 15 and 20 spectral bands, approaching hyperspectral resolutions of sensors. We hypothesize that the general NCA is capable of learning and simulating persistent target images at higher spectral resolutions. The second experiment (*towards virtual inoculation*) investigates the signal-responding NCA variant [27] for transforming an existing healthy leaf section into pathogen-induced disease stages consistently across 10 spectral bands. The NCA responds to *internal* signals representing genetically defined infection stages and an *external* signal simulating virtual inoculation. Based on fixed regularity capturing disease progression, this experiment serves as a proof-of-concept for controllably evolving healthy leaf tissue into a specified target disease stage. Hyperparameter optimization (HPO) was performed using the Optuna framework with the Tree-structured Parzen Estimator (TPE) sampler [1], limited to 100 trials (number of trained NCAs) each. In the *first experiment*, 100 trials were conducted for each of the $n = 5$ spectral resolution, separately performed for healthy and infected target leaf sections; in the *second experiment*, 100 trials were conducted each of the signal initialization strategies (*single cell* and *complete grid*). Table 1 provides an overview of the varied hyperparameters. Specific

Table 1: Search space of Optuna hyperparameter optimization.

Hyperparameter	Experiment 1	Experiment 2
Cell state channels	[16, 35]	[19, 35]
Cell fire rate	[0.10, 0.90]	[0.10, 0.90]
Target padding	{25, 30, ..., 50}	{10, 15, 20}
Batch size	{4, 8}	{8, 12}
Pool size	[1000, 1500]	[1000, 1500]
Seed cell replacement	{1, 2}	{2, 3}
Learning rate	$[10^{-4}, 5 \times 10^{-3}]$	$[10^{-4}, 5 \times 10^{-3}]$
Learning rate schedule step	{10000, 15000}	{20000, 25000}

adjustment of the initialization of the external stimulus, refer to the number of cells receiving the signal during the signaling training strategy. The signal is applied to a central *single cell* within the grid of leaf target section, as intended in the introduction of the signaling NCA variant. In the recently proposed signaling NCA variant [27], the signal is applied to a *single cell*. In our experiment this is expanded to applying the trigger to the *complete grid* at once. Simulating a real inoculation in a wet laboratory experiment. Not varied hyperparameters include: The training duration which is fixed to 20,000 epochs in the first experiment, and in the second experiment increased to 30,000 epochs, unless early termination criteria were met (cf. Supp. Mat. 1.9 for additional details regarding the training and early stopping). Both experiments were repeated $N = 30$ times using the most promising hyperparameter configuration identified by Op-

tuna. In the first experiment, repetitions cover each spectral resolution for both healthy and infected targets; in the second, both signal initialization strategies. The NCA organism with the *structural fidelity* measured by the Universal Image Quality Index (UIQI) [32] closest to the ensemble mean across all repetitions was selected as the representative organism for qualitative evaluation and computational efficiency analysis. Evaluation was conducted during *inference* (after training the NCA variants cf. Sect. 3), growing or transforming a target leaf sections from a seed cell applying the learned update rule for 300 steps. In the first experiment, the NCA grows from a single seed cell; in the second, it transforms from a healthy leaf state. *Quantitative evaluation* assesses both *structural fidelity* - maintaining the visual structure of the target leaf section - and *spectral fidelity* - reconstructing the spectral signature of multispectral images - of the NCA-simulated images. Prior to metric computation, simulated images are rescaled to their physical reflectance range based on global ground truth (GT) target minimum and maximum values. Structural fidelity is evaluated for each of the $n = 5$ spectral resolution per spectral band and averaged across all valid leaf pixels using three metrics: the Root Mean Squared Error (RMSE), measures the average pixel-wise prediction error in physical reflectance units; the Structural Similarity Index (SSIM) [33], ranging from -1 to 1 where 1 indicates perfect agreement; and the UIQI [32], equally ranging from -1 to 1 . Spectral fidelity, evaluated only for $n = 4$ multispectral resolutions, is assessed pixel-wise across the spectral dimension and averaged over all valid pixels using the Spectral Angle Mapper (SAM) [35], where values closer to 0 radians (rad) indicate higher fidelity. Furthermore, the Spectral Pearson Correlation Coefficient (SPCC) [30], where 1 indicates perfect agreement is evaluated. *Computational efficiency* is assessed based on trainable parameter counts of the representative organism. Training was conducted on locally on a GPU server with an A100-SXM4-80GB (MIG partition) and CPU (inference) on a AMD Ryzen 7 7840HS. *Qualitative evaluation* compares NCA-simulated images visually against the GT using pseudo-RGB images or band-wise grayscale representations for multispectral resolutions.

4.1 Results of Experiment 1

The quantitative performance metrics, involving structural fidelity of the NCA simulated images across RGB, and multispectral resolutions and representative parameter counts are displayed in Table 2. The performance refers to the NCA simulation capability. For the infected target leaf section, highest performance is achieved for the RGB baseline, and lowest at 20 bands. SSIM and UIQI performance is highest for the RGB baseline, and decrease with higher spectral resolution, with lowest performance at 20 bands. The healthy leaf section target shows highest RMSE performance for the RGB baseline, decreasing in performance when scaling to spectral resolution. Similar as with the infected target, both SSIM and UIQI reach highest performance at the RGB baseline. Resulting in parameter counts for the representative organisms, $12\text{ K} - 18\text{ K}$ depending on the spectral resolution (number of channels C_i , C_h) lowest at the RGB baseline,

Table 2: Quantitative results of experiment 1 (spectral resolution scaling), RMSE, SSIM and UIQI values as means ± 1 SD over $N = 30$ repetitions; bold values indicate best performance. Parameter counts θ refer to the representative organism.

Target	Bands	RMSE $\downarrow$	SSIM $\uparrow$	UIQI $\uparrow$	θ
<i>Infected</i>	RGB	0.0032 ± 0.0008	0.9314 ± 0.0260	0.9192 ± 0.0260	12.4 K
	5	0.0075 ± 0.0028	0.9204 ± 0.0359	0.8934 ± 0.0433	15.5 K
	10	0.0075 ± 0.0211	0.9268 ± 0.0211	0.8885 ± 0.0363	18.0 K
	15	0.0077 ± 0.0008	0.9032 ± 0.0143	0.8752 ± 0.0175	18.0 K
	20	0.0096 ± 0.0001	0.8602 ± 0.0141	0.8188 ± 0.0211	17.5 K
<i>Healthy</i>	RGB	0.0026 ± 0.0004	0.9316 ± 0.0142	0.9277 ± 0.0149	11.9 K
	5	0.0045 ± 0.0012	0.9486 ± 0.0211	0.9264 ± 0.0482	14.5 K
	10	0.0051 ± 0.0013	0.9427 ± 0.0181	0.9082 ± 0.0324	14.5 K
	15	0.0052 ± 0.0004	0.9322 ± 0.0061	0.9094 ± 0.0098	17.0 K
	20	0.0059 ± 0.0004	0.9193 ± 0.0059	0.8864 ± 0.0093	18.0 K

with slightly lower parameter counts for the healthy target. Overall, a higher performance can be observed in the healthy target when compared to the infected leaf section across all spectral resolutions. For the SPCC, high values across all spectral resolutions indicate a near-perfect reconstruction of the GT spectral signatures. Likewise, low values for the SAM can be observed across all resolutions. This indicates that the spectral signature can be successfully maintained when increasing the spectral resolution. Overall, similar to the structural metrics, highest performance could be observed in reconstructing the healthy target, with best values at 10 spectral bands. The spectral band-wise results indicate non-uniform performance across spectral resolutions, performance declines are observed at the outer spectral bands (cf. Supp. Mat. 1.10). Figure 2 displays the NCA-simulated spectral signatures of the representative organisms against the GT for all spectral resolutions, showing mean ± 1 SD and the 95th percentile range. Note that confidence intervals are omitted as their assumption of independent observations is violated due to assumed spatial autocorrelation between neighboring pixels. At all resolutions, the simulated mean spectra closely reconstruct the GT for both targets. While blue bands (444–475 nm) exhibit low reflectance variability, the NIR range (705–940 nm) shows higher variability, where the NCA simulations reveal a slight decline in capturing the full spread, visible in deviations from the 95th percentile range and pixel-wise extrema.

Besides assessing quantitative performance, the following exemplary qualitative evaluation in Figure 3 displays representative NCA organisms, as simulated leaf section compared to the GT, both as pseudo-RGB and 10 spectral band resolutions as grayscale images for the infected target. The NCA tended to successfully reconstruct the biological structure of both the leaf and the pathogen-induced tissue at both RGB and multispectral resolutions, rather than the physical sensor noise (visible at 444 nm).

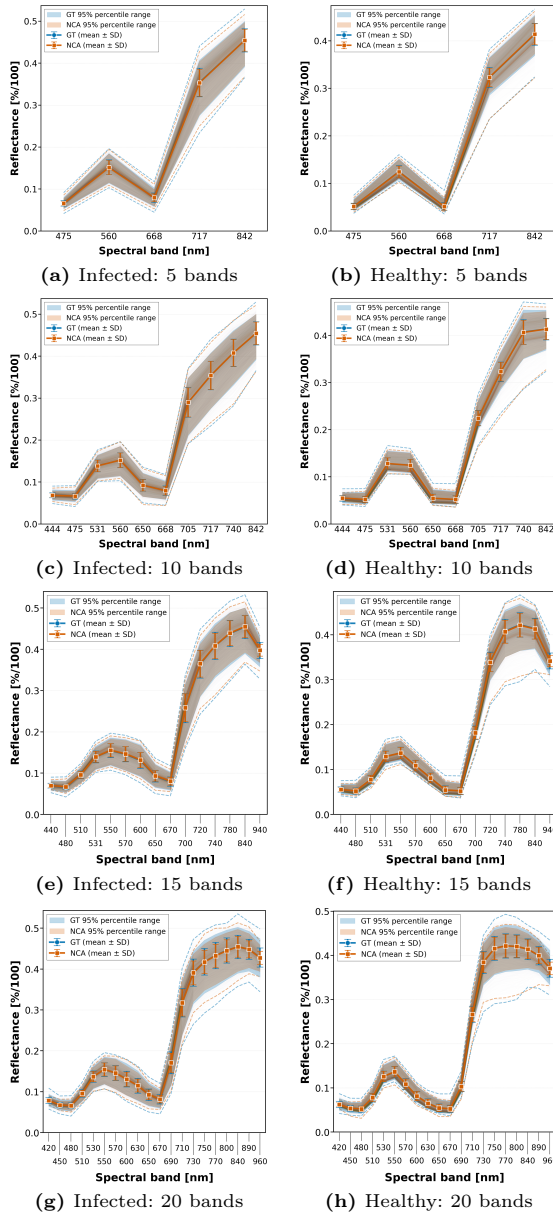


Fig. 2: Comparison of spectral signatures of experiment 1 at 5, 10, 15, and 20 bands for infected and healthy multispectral resolution targets.

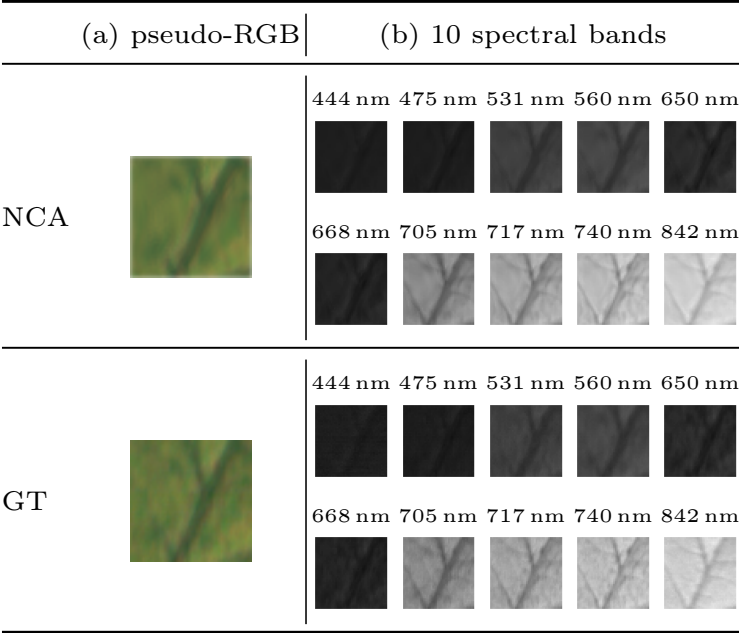


Fig. 3: Qualitative spectral band-wise comparison of NCA and GT of experiment 1: (a) pseudo-RGB and (b) 10-band grayscale images for the infected target.

4.2 Results of Experiment 2

Table 3 shows the quantitative results and representative parameter counts of the structural fidelity of the signaling-responding NCA simulated images for the two distinct stages progressive in pathogen infection. Performance is measured by the RMSE, SSIM, and UIQI averaged over 10 spectral bands. The evaluation of structural fidelity in the second experiment needs to be distinguished: *First*, regarding the difference in performance in both the earlier infection target (6 DAI) and the more symptomatic infection target (8 DAI), where notably the later one achieved higher performance overall when taking into account all structural metrics. *Second*, the performance can be evaluated comparing the two signaling initialization methods; applying the external signal to a single cell, or the entire cell grid during training. The strategy using a single cell slightly outperformed the approach of using the complete grid in the earlier infection stage (6 DAI). Interestingly, in the more symptomatic infection stage (8 DAI) target, both signal initialization strategies achieved near identical performance. Resulting in parameter counts for the representative organisms, ranging 14K–15K, where both signal initialization strategies achieved similar values. Similarly, the more symptomatic infection target (8 DAI) achieved higher spectral fidelity, measured by SAM and SPCC, averaged over all pixels displayed in Table 4. Taking into account the signal initialization strategy, both variants achieved similar results, with the complete grid strategy achieving slightly higher spectral

Table 3: Quantitative results of experiment 2 (towards virtual inoculation) for 10 spectral bands, RMSE, SSIM and UIQI as means ± 1 SD over $N = 30$ repetitions; bold values indicate best performance. Parameter counts θ refer to the representative organism.

Signal	Target	RMSE $\downarrow$	SSIM $\uparrow$	UIQI $\uparrow$	θ
		mean ± 1 SD	mean ± 1 SD	mean ± 1 SD	
Single cell	<i>Infected (6 DAI)</i>	0.0034 $\pm$ 0.0005	0.9638 $\pm$ 0.0067	0.9680 $\pm$ 0.0441	14.8 K
	<i>Infected (8 DAI)</i>	0.0025 $\pm$ 0.0003	0.9850 $\pm$ 0.0029	0.9948 $\pm$ 0.0044	
Complete grid	<i>Infected (6 DAI)</i>	0.0051 $\pm$ 0.0013	0.9487 $\pm$ 0.0097	0.9480 $\pm$ 0.0622	13.8 K
	<i>Infected (8 DAI)</i>	0.0025 $\pm$ 0.0004	0.9847 $\pm$ 0.0037	0.9949 $\pm$ 0.0042	

fidelity. Again, these results confirm that the NCA-simulated images successfully reconstruct the specific spectral signature, in this case of two distinct stages of pathogen-infection. Note that the difference in performance of the initialization

Table 4: Quantitative evaluation of experiment 2 (towards virtual inoculation), SAM and SPCC across for 10 spectral bands resolutions, as means ± 1 SD over $N = 30$ repetitions; bold values indicate the best performance per target.

Signal	Target	SAM (rad) $\downarrow$	SPCC $\uparrow$
		mean ± 1 SD	mean ± 1 SD
Single cell	<i>Infected (6 DAI)</i>	0.0111 $\pm$ 0.0004	0.9998 $\pm$ 0.0001
	<i>Infected (8 DAI)</i>	0.0094 $\pm$ 0.0002	0.9999 $\pm$ 0.0001
Complete grid	<i>Infected (6 DAI)</i>	0.0144 $\pm$ 0.0013	0.9997 $\pm$ 0.0002
	<i>Infected (8 DAI)</i>	0.0093 $\pm$ 0.0003	0.9999 $\pm$ 0.0001

methods, either applied through a single cell or to the entire grid, could be evaluated statistically. However, the main contribution of this paper is providing a proof-of-principle of our scenarios for implementing general NCA variants, including the extension of the signal initialization in the signaling variant [27] to the entire grid. Therefore, a detailed statistical evaluation falls beyond the scope of this initial work and is planned to be conducted in future work.

For additional qualitative evaluation Figure 4, display the transformation from the healthy leaf sections into pathogen-infected stages represented as pseudo-RGBs. The region of signal initialization (cf. Sect. 3), as well as the propagation of the signal across the cell grid over a range of 100 steps during the inference is visualized for qualitative assessment of our simulated virtual inoculation.

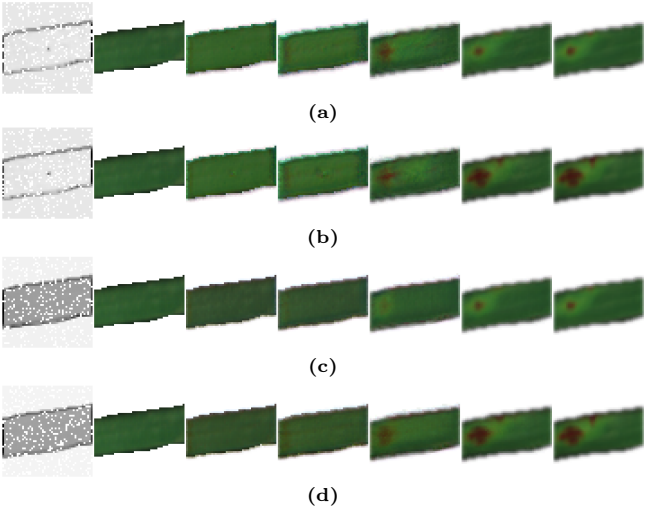


Fig. 4: Qualitative comparison of signal initialization of experiment 2: single cell for (a) early (6 DAI) and (b) late (8 DAI) stage; complete grid for (c) early (6 DAI) and (d) late (8 DAI) stage, showing pseudo-RGB initialization and progression over inference steps upon signal activation (C_E).

5 Discussion

Overall, the *first experiment* confirmed our hypothesis that the NCA can successfully simulate multispectral images, as demonstrated by the high structural and spectral fidelity achieved. At lower spectral resolutions (10 bands), the NCA accurately reconstructed fine-scaled, biologically tissue-specific structural variations. At higher resolutions (15 and 20 bands), a decrease in fine-scaled structural performance could be observed (cf. Table 2). This is directly attributed to an architectural bottleneck; the number of cell states was increased to a maximum of $C = 35$, expanding the perception vector to 105 ($3C$) features for each cell. Processing higher-dimensional features through a single hidden layer constrained to only $128\ 1 \times 1$ kernels, is hypothesized to have lead the NCA to prioritize dominant spectral signatures over fine-grained structural details. However, the pixel-wise spectral fidelity remained high and consistent across all spectral resolutions. The second experiment involved extending the simulation from the growth of a static target to the transformation of an existing healthy leaf into pathogen-induced stages Moving beyond the reconstruction of a static target towards enabling the NCA for virtual inoculation of healthy plant leaves. This successful transformation suggests that the signalling NCA variant can learn multiple disease stages simultaneously across ten spectral bands, indicating its potential for modelling more temporally resolved disease progression.

Limitations Notably, the scope of the initial experiments conducted in this paper were confined to the reconstruction of a predefined leaf section, or the transfor-

mation of an existing leaf section into a predefined subsequent stage of pathogen-infection. It does not support the learning of statistical distribution regarding how disease symptoms might appear on the crop leaves. Consequently, the experiments presented herein lack the capabilities of state-of-the art generative models to be used as - true - data augmentation techniques. That said, it was not the objective of this proof-of-concept to employ generative NCA variants (cf. Sect. 2), but is already planned in further work. More advanced qualitative metrics and model parameter-counts can then be directly compared between spectrally-scaled Hybrid-NCA-architectures [7, 10], synthesizing images of higher-resolution (beyond 50×50 pixels) and state-of-the art generative AI models. A direct comparison was not performed in the presented experiments, since it is not possible to make a valid assertion on the matter. However, it remains to be seen whether these NCA variants can simulate high-spatial and spectral coherence in high-resolution images of pathogen-infected plant leaves. Furthermore, it should be noted that the experiments were restricted to two specific combinations of plant pathogens. Similarly, the implementation of sensor-specific band resampling was constrained by the limitations of available hardware. It is possible that the simulated images will demonstrate varying degrees of performance when scaled to different spectral bands than those chosen in the experiments.

6 Conclusion

We studied and provided proof-of-principle of NCAs as self-organizing systems approach, to simulate data, in the context of crop disease prediction. Initial implementations demonstrated successful scaling to higher spectral resolutions, resembling state-of-the-art multispectral imaging systems, while maintaining high structural and spectral fidelity at a substantially lower parameter count of only $\approx 18\text{K}$ or less. Our results suggest that NCAs show further potential for resource-efficient augmenting plant-pathogen datasets in automated disease detection and integration into advanced sensor systems. Overall these findings further demonstrate the potential of employing NCAs for simulating high resolution data, offering a bio-inspired and inherently interpretable simulation of synthetic image data.

Future research will expand the scope of these initial experiments through the implementation of generative Hybrid-NCA variants, aimed at facilitating the simulation of full-resolution pathogen-infected crop leaf images. This paves the way to further explore whether simulating spatially-spectrally highly-resolved disease-progression is feasible. In addition or proposed concept of virtual inoculation will be explored in greater depth, aiming to extend or method towards the generative transition between distinct disease stages. Further work will address possible VRAM memory and training time limitations when extending to Hybrid-NCA variants, as well as the comparison to state-of-the-art generative AI models. The objective is to maintain the viability of these models for multispectral agricultural applications on resource-constrained hardware, thereby contributing to the mitigation of data scarcity in plant disease detection.

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